

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination – 2024

Course: B. Tech. Branch: Mechanical Engineering /Mechanical Engineering (Sandwich)

Semester :VII

Subject Code & Name: Non- conventional Machining (BTMPE703C)

Max.Marks: 60 Date: 10/02/2025 Duration: 3 Hr.

Instructions to the Students:
1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Mark
Q. 1	Objective type questions. (Compulsory Question)		12
1	Non-Traditional machining can also be called as?	Recall/CO1	1
	a. Contact machining c. Partial contact machining b. Non-contact machining d. Half contact machining		
2	Which of the following is a conventional process of machining?	Recall/CO1	1
	a. Cylindrical Grinding c. Electron beam machining b. Electrochemical machining d. Water jet machining		
3	In electro-discharge machining, dielectric is used to -----	Recall/CO2	1
	a. help in the movement of the sparks c. act as coolant b. control the spark discharges d. all of the mentioned		
4	In the laser beam machining process, as compared to other unconventional machining processes, the heat-affected zone is -----	Recall/CO2	1
	a. Larger c. Not present b. Smaller d. Moderate		
5	Ultrasonic Machining can be used for which of the following processes and applications?	Recall/CO3	1
	a. Drilling c. Polishing b. Sinking and Contouring d. All of the above		
6	Which of the following material cannot be machined using ECM?	Understand /CO3	1
	a. High strength alloys c. Non conductive materials b. Hardened steel d. All of the above		
7	What are the machining rates used in Micro-EDM process?	Recall/CO6	1
	a. 0.1 to 2 mm/min c. 10 to 200 mm/min b. 1 to 20 mm/min d. 100 to 500 mm/min		
8	Which of the following is an example of hybrid machining?	Recall/CO5	1
	a. Ultrasonic Machining c. Ultrasonic assisted electrochemical machining b. Electron Beam Machining d. Laser Beam Machining		

9	Which type of materials cannot be machined using Abrasive jet machining?		Recall/CO4	1
	a. Soft materials b. Hard materials	c. Difficult to cut materials d. None of the above		
10	How are the machining times in Laser beam machining?		Recall/CO4	1
	a. Very slow b. Slow	c. Moderate d. Fast		
11	In ECG, grinding action removes _____		Understand /CO5	1
	a. workpiece material from the surface b. corroded surface layer making workpiece ready for electro-chemical process	c. oxide film formed on the surface d. particles removed from surface after chemical action		
12	Which of the following is true about AJMM?		Recall/CO6	1
	a. It is a low cost method b. It is toxic towards the environment	c. It is hazardous to human health d. It is not capable of anisotropic machining		
Q. 2 Solve the following.				12
A)	Discuss the factors to be considered while selecting the non-conventional machining processes.		Understand /CO1	6
B)	Explain the classification of unconventional machining according to major energy source employed. What is the importance of unconventional machining?		Understand /CO1	6
Q.3 Solve the following.				12
A)	Explain the different types of maskants and etchants used in chemical machining process.		Understand /CO2	6
B)	Explain with a neat sketch the working principle and the parametric influence of abrasive jet machining.		Understand /CO3	6
Q. 4 Solve Any Two of the following.				12
A)	What are the main functions of electrolysis in the ECM? How the current density affects the material removal rate?		Understand /CO3	6
B)	List the various process parameters of USM and explain the effect of amplitude, frequency, static load and abrasive size in MRR and surface finish.		Recall/CO3	6
C)	Explain process parameters and process characteristics of PAM process.		Understand /CO3	6
Q.5 Solve Any Two of the following.				12
A)	List the merits and demerits of Laser Beam Machining process.		Recall/CO4	6
B)	Differentiate between the water jet cutting and the abrasive water jet cutting.		Understand /CO4	6
C)	Explain with a neat diagram the working principle of abrasive jet machining and its applications.		Understand /CO4	6
Q. 6 Solve Any Two of the following.				12
A)	Sketch and explain the electro-chemical Honing process.		Understand /CO5	6

B)	What is the process of ELID Grinding? Differentiate between conventional grinding with the ELID Grinding.	Understand /CO5	6
C)	How internal surface of components can be deburred, polished by abrasive flow finishing? Explain.	Understand /CO5	6
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